

System Hardening Report

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Summary of Flags

The following security configuration flags were identified during the system-hardening assessment:

FLAG_USER_TALA_NOT_IN_SUDO

FLAG_GROUP_VOYAGERS_CREATED

FLAG_5.2.13_STRONG_CIPHERS_USED

FLAG_5.2.14_STRONG_MACS_USED

FLAG_5.4.1_PASSWORD_POLICY_CORRECT

FLAG_5.2.7_ROOT_LOGIN_DISABLED

FLAG_5.2.21_LOGIN_GRACE_TIME_CORRECT

FLAG_2.2.8_FTP_NOT_INSTALLED

FLAG_2.2.9_HTTP_NOT_INSTALLED

FLAG_2.2.3_CUPS_NOT_INSTALLED

FLAG_3.5.1.3_UFW_ACTIVE

FLAG_3.5.1.7_DEFAULT_FIREWALL_POLICY_CORRECT

FLAG_3.5.1.6_FIREWALL_RULES_EXIST

FLAG_USER_KAKAMORA_REMOVED

FLAG_TAMATOA_PASSWORD_CHANGED

Total flags identified: 15

Vulnerabilities Identified

Vulnerability 1: Excessive sudo privileges

The sudo group initially included unauthorized users. In particular, **Tala** had administrative privileges even though the company's policy limited privileged access to authorized administrators.

Evidence from the system showed Tala as a member of the sudo group.

After remediation, the sudo group was verified as:

```
sudo:x:27:Moana,Tamatoa,Maui,Tui
```

Tala was successfully removed.

Security impact: Unauthorized administrative access could allow a user to modify system configurations, access sensitive information, or disable security controls.

Vulnerability 2: Unauthorized user account

The Kakamora account was present on the workstation even though it was not an authorized user.

The initial audit showed:

```
uid=1008(Kakamora) gid=1008(Kakamora) groups=1008(Kakamora)
```

The account was removed using userdel.

After remediation:

```
id: 'Kakamora': no such user
```

Security impact: Unnecessary accounts increase the attack surface and could provide an unauthorized path into the system.

Vulnerability 3: Incorrect Voyager group configuration

The required voyagers group was not initially configured correctly.

The group was created and configured with the required members:

```
voyagers:x:1009:Moana,Pua,Heihei
```

Security impact: Incorrect group membership could prevent authorized users from accessing required resources or allow unauthorized users to access them.

Vulnerability 4: Weak SSH configuration

The original SSH configuration allowed:

```
permitrootlogin yes
```

```
logingracetime 120
```

```
ciphers aes128-cbc,3des-cbc
```

```
macs hmac-md5,hmac-sha1
```

These settings presented several security weaknesses.

Root login was enabled, the login grace period was too long, and outdated cryptographic algorithms were being used.

The configuration was hardened to:

```
permitrootlogin no
```

```
logingracetime 60
```

and modern encryption ciphers and MAC algorithms were configured.

Security impact: Weak SSH configurations can increase the likelihood of unauthorized remote access, credential attacks, and exploitation of outdated cryptographic algorithms.

Vulnerability 5: Unnecessary network services installed

The workstation initially contained:

- vsftpd
- apache2
- cups

These services were not required for the workstation's intended purpose.

They were removed from the system.

Verification showed that vsftpd and cups were no longer installed and Apache was marked as uninstalled.

Security impact: Unnecessary services increase the attack surface and can expose additional ports and vulnerabilities.

Vulnerability 6: Firewall was initially disabled

The initial UFW status was:

Status: inactive

This meant the workstation did not have the required firewall protection.

UFW was enabled and configured with:

Default: deny (incoming), allow (outgoing), deny (routed)

SSH access was explicitly allowed on port 22.

Security impact: A disabled firewall can allow unwanted network traffic to reach services running on the workstation.

Vulnerability 7: Password and account security concerns

The assessment also reviewed password settings and account aging.

The system's configured password policy included:

PASS_MAX_DAYS 1

PASS_MIN_DAYS 0

PASS_WARN_AGE 7

PAM was configured to use pam_unix with yescrypt.

The assessment also verified Tamatoa's password status:

Tamatoa P 12/19/2024 0 99999 7 -1

The lab identified the Tamatoa password-change requirement as satisfied.

Remediation Strategies

Remediation 1: Restrict administrative privileges

Remove unauthorized users from the sudo group and review privileged accounts regularly.

The final sudo group was:

Moana,Tamatoa,Maui,Tui

This follows the company's requirement that only authorized administrators have privileged access.

Ongoing recommendation: Conduct a quarterly review of privileged accounts.

Remediation 2: Remove unauthorized accounts

The Kakamora account was removed.

Going forward, DataTrust should implement an account-management process requiring approval before accounts are created and requiring immediate removal when users leave the organization.

Remediation 3: Manage group membership

The voyagers group was created with the required members:

Moana,Pua,Heihei

Group memberships should be reviewed regularly to ensure employees only have access required for their responsibilities.

Remediation 4: Harden SSH

SSH was configured to:

- Disable direct root login.
- Reduce the login grace period.
- Remove weak ciphers.
- Remove weak MAC algorithms.
- Allow only stronger cryptographic algorithms.

The final configuration was verified using `sshd -T`.

Ongoing recommendation: Review SSH configuration after major operating-system or OpenSSH updates.

Remediation 5: Remove unnecessary services

FTP, HTTP, and CUPS packages were removed because they were not required.

This reduces the system's attack surface and makes future security monitoring easier.

Remediation 6: Enable and configure UFW

UFW was enabled with a default-deny incoming policy and default-allow outgoing policy.

SSH was explicitly permitted:

22/tcp ALLOW IN Anywhere

22/tcp (v6) ALLOW IN Anywhere (v6)

This provides basic network-level protection while maintaining required remote administration.

Hardening Process Summary

Overview of Hardening Process

The hardening process began by auditing the workstation to identify users, groups, password settings, SSH configuration, installed services, and firewall status. The identified weaknesses were then remediated according to the company's security requirements.

Steps Taken

1. Verified the operating system was Ubuntu 22.04.
2. Audited user accounts and group memberships.
3. Reviewed administrator privileges.
4. Removed unauthorized Kakamora account.
5. Removed Tala from the sudo group.
6. Created and configured the voyagers group.
7. Reviewed password policy and account settings.
8. Audited SSH configuration.
9. Disabled SSH root login.
10. Reduced SSH login grace time.
11. Replaced weak SSH ciphers and MACs.

12. Removed unnecessary FTP, HTTP, and CUPS services.
13. Enabled UFW.
14. Configured default firewall policies.
15. Added an SSH firewall rule.
16. Verified the final security configuration.

Tools Used

The main tools used included:

- `id`
- `getent`
- `passwd`
- `chage`
- `sshd`
- `dpkg`
- `apt`
- `ufw`
- `ss`
- Linux configuration files and terminal utilities

Challenges and Observations

One challenge was that the Docker environment did not have nano installed. The SSH configuration was therefore modified using command-line tools instead.

Another observation was that `systemctl` could not be used because the assessment container was not running `systemd` as its initialization system. Package verification and other available Linux commands were used instead.

The initial SSH configuration contained weak cryptographic settings and allowed root login. These settings were successfully hardened and verified.

Conclusion

The system-hardening assessment identified several security weaknesses involving privileged accounts, unauthorized users, SSH configuration, unnecessary services, and firewall protection.

The workstation was successfully hardened by removing unauthorized access, configuring the required user groups, removing unnecessary services, strengthening SSH, and enabling UFW with a default-deny incoming policy.

The final configuration significantly improves the workstation's security posture and better aligns it with the company's security requirements.

Continued security should include regular user and group reviews, periodic SSH and firewall audits, timely software updates, password-policy reviews, and monitoring of installed services. Maintaining these practices will help prevent configuration drift and reduce the likelihood of unauthorized access or other security incidents.

Overall, the workstation's security posture was substantially improved through the hardening process, with 15 assessment flags identified and addressed.

```
Moana@b7ccableffe0:~$ getent passwd | cut -d: -f1
root
daemon
bin
sys
sync
games
man
lp
mail
news
uucp
proxy
www-data
backup
list
irc
gnats
nobody
_apt
systemd-network
systemd-resolve
messagebus
systemd-timesync
syslog
ftp
sshd
avahi
saned
colord
Moana
Tamatoa
```

```
news
uucp
proxy
www-data
backup
list
irc
gnats
nobody
_apt
systemd-network
systemd-resolve
messagebus
systemd-timesync
syslog
ftp
sshd
avahi
saned
colord
Moana
Tamatoa
Maui
Tui
Heihei
Pua
Tala
Sina
Kakamora
```

```
Moana@b7ccableffe0:~$ getent group voyagers
```

```
Moana@b7ccableffe0:~$ getent group voyagers
```

```
Moana@b7ccableffe0:~$ sudo -l
```

```
Matching Defaults entries for Moana on b7ccableffe0:
```

```
env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin, use_pty
```

```
User Moana may run the following commands on b7ccableffe0:
```

```
(ALL : ALL) ALL
```

```
(ALL) NOPASSWD: ALL
```

```
Moana@b7ccableffe0:~$ SS
```

```
Moana@b7ccableffe0:~$ id Maui
uid=1002(Maui) gid=1002(Maui) groups=1002(Maui),27(sudo)
Moana@b7ccableffe0:~$ id Tui
uid=1003(Tui) gid=1003(Tui) groups=1003(Tui),27(sudo)
Moana@b7ccableffe0:~$ id Tamatoa
uid=1001(Tamatoa) gid=1001(Tamatoa) groups=1001(Tamatoa),27(sudo)
Moana@b7ccableffe0:~$ id Heihe
id: 'Heihe': no such user
Moana@b7ccableffe0:~$ id Pua
uid=1005(Pua) gid=1005(Pua) groups=1005(Pua)
Moana@b7ccableffe0:~$ id Tala
uid=1006(Tala) gid=1006(Tala) groups=1006(Tala),27(sudo)
Moana@b7ccableffe0:~$
```

```
Moana@b7ccableffe0:~$ getent group voyagers
voyagers:x:1009:
```

```
Moana@b7ccableffe0:~$ getent group voyagers
voyagers:x:1009:Moana,Pua,Heihe
```

```
Moana@b7ccableffe0:~$ sudo deluser Tala sudo
Removing user 'Tala' from group 'sudo' ...
Done.
Moana@b7ccableffe0:~$ getent group sudo
sudo:x:27:Moana,Tamatoa,Maui,Tui
```

```
Moana@b7ccableffe0:~$ sudo chage -l Moana
Last password change                : Dec 19, 2024
Password expires                    : never
Password inactive                   : never
Account expires                     : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
Moana@b7ccableffe0:~$ sudo chage -l Maui
Last password change                : Dec 19, 2024
Password expires                    : never
Password inactive                   : never
Account expires                     : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
^[[200~b7ccableffe0:~$ ^C
Moana@b7ccableffe0:~$ sudo chage -l Tui
Last password change                : Dec 19, 2024
Password expires                    : never
Password inactive                   : never
Account expires                     : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
Moana@b7ccableffe0:~$ sudo chage -l Tamatoa
Last password change                : Dec 19, 2024
Password expires                    : never
Password inactive                   : never
Account expires                     : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
```

```
Moana@b7ccableffe0:~$ grep -E 'PASS_MAX_DAYS|PASS_MIN_DAYS|PASS_WARN_AGE' /etc/login.defs
# PASS_MAX_DAYS Maximum number of days a password may be used.
# PASS_MIN_DAYS Minimum number of days allowed between password changes.
# PASS_WARN_AGE Number of days warning given before a password expires.
```

```
PASS_MAX_DAYS 1
PASS_MIN_DAYS 0
PASS_WARN_AGE 7
```

```
Moana@b7ccableffe0:~$ grep -E 'pam_pwquality|pam_unix' /etc/pam.d/common-password
# used to change user passwords. The default is pam_unix.
# Explanation of pam_unix options:
# 'OBSOLETE_CHECKS_ENAB' option in login.defs. See the pam_unix manpage
password [success=1 default=ignore] pam_unix.so obscure yescrypt
auth required pam_unix.so sha512
```

```
Moana@b7ccableffe0:~$ sudo sshd -T | grep -E 'permitrootlogin|loggingracetime|ciphers|macs'
loggingracetime 120
permitrootlogin yes
ciphers aes128-cbc,3des-cbc
macs hmac-md5,hmac-sha1
```

```
Moana@b7ccableffe0:~$ sudo ss -tulpn | grep ssh
tcp    LISTEN 0      128          0.0.0.0:22    0.0.0.0:*    users:(("sshd",pid=184,fd=3))
tcp    LISTEN 0      128          [::]:22      [::]:*       users:(("sshd",pid=184,fd=4))
```

```
Moana@b7ccableffe0:~$ sudo sshd -T | grep -E 'permitrootlogin|loggingracetime|ciphers|macs'
loggingracetime 120
permitrootlogin yes
ciphers aes128-cbc,3des-cbc
macs hmac-md5,hmac-sha1
```

```
Moana@b7ccableffe0:~$ sudo sshd -T | grep -E 'permitrootlogin|loggingracetime|ciphers|macs'
loggingracetime 60
permitrootlogin no
ciphers chacha20-poly1305@openssh.com,aes256-gcm@openssh.com,aes128-gcm@openssh.com,aes256-ctr,aes192-ctr,aes128-ctr
```

```
Moana@b7ccableffe0:~$ sudo ufw allow 22/tcp
```

Rule added

Rule added (v6)

```
Moana@b7ccableffe0:~$ sudo ufw status numbered
```

Status: active

	To	Action	From
	--	-----	----
[1]	22/tcp	ALLOW IN	Anywhere
[2]	22/tcp (v6)	ALLOW IN	Anywhere (v6)

```
Moana@b7ccableffe0:~$ getent group voyagers
```

```
voyagers:x:1009:Moana,Pua,Heihei
```

```
Moana@b7ccableffe0:~$ voyagers:x:...:Moana,Pua,Heihei
```

```
Moana@b7ccableffe0:~$ getent group voyagers
```

```
voyagers:x:1009:Moana,Pua,Heihei
```

```
Moana@b7ccableffe0:~$ voyagers:x:...:Moana,Pua,Heihei
```

```
bash: voyagers:x:...:Moana,Pua,Heihei: command not found
```